

Daein Weon

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RESEARCH INTERESTS

Evaluation and measurement for machine-learning systems: what a reported score entitles a practitioner to claim, and how a system should act once a measurement says something has gone wrong. My work so far asks where benchmark scores, proxy labels and model-based judges stop supporting the claims drawn from them — in safety routing, tool-using agents and reasoning verifiers — and builds inference-time controls that act on those measurements. I would like to extend this to multi-agent and reasoning systems, to evaluation that has to support a decision rather than a ranking, and to using the same diagnostics during training rather than only after it.

EDUCATION

Kookmin University, Seoul, South Korea

Mar 2019 – Feb 2027 (expected)

B.B.A. in AI Big Data Analytics and Statistics; B.S. in Computer Science (double major)

GPA 3.74 / 4.5

PUBLICATIONS

1. From Evidence to Exact Answers: Copy-Preserving Compression and Conversion-Aware Calibration for Frozen SLM RAG.

Dong Ho Kang, Hyeonjeong Cha, Daein Weon.

Proceedings of EMNLP 2026 (Main Conference). Accepted August 2026.

Evidence-exposure and answer-transformation levers for retrieval with frozen small language models: a pointer-preserving card compressor and an (interface, prompt) calibrator, evaluated on five LongBench tasks with three 1.2–4B readers.

UNDER REVIEW

1. Same Trajectories, Different Credit: Scoring-Rule Dependence in Tool-Fault Recovery Evaluation.
Submitted to ICLR 2027.

A tool-fault benchmark ships two rules for which calls an agent may still make after a failure and scores with the stricter one. Re-scoring identical stored trajectories under scorers differing in that one condition moves the pass rate by 11–47 points and halves the gain credited to a recovery instruction; a pre-registered replication reproduces the first effect and not the second.

2. What a Reported Pooled AUC Identifies When Scores Are Unavailable.

Submitted to ICLR 2027.

In stratified evaluations a pooled AUC has no single chance level. Sharp bounds on its conditional permutation mean are derived and shown attainable, and computed from labels alone on public benchmarks, where the admissible interval spans most of the scale.

3. Knowledge-Based Zero-Replay Debugging of Multi-Agent LLM Traces.

Hyeonjeong Cha, Daein Weon, Dong Ho Kang.

Submitted to AAMAS 2027. Preprint: arXiv:2606.14805.

Counterfactual replay localises the decisive events in a multi-agent execution trace but costs one re-run per candidate event. Each trace is compiled into an event knowledge graph and a calibrated ranker predicts which events a replay oracle would mark high-effect, raising held-out localisation from 0.73 to 0.93 Branch Recall@5 at zero replay cost.

MANUSCRIPTS IN PREPARATION

1. The Attribution Contract: Why Step Accuracy Is Not Enough.

To be submitted to AAMAS 2027.

Failure-attribution benchmarks are audited as a score–proxy–construct contract and fail at every link; the contract is then meta-validated on an injected-failure testbed whose labels are verified by re-execution.

PREPRINTS

1. Auditing Proxy-Based Validation Across Text Spans.

Daein Weon, Dong Ho Kang.

arXiv:2609.25808, September 2026.

Agreement between an evaluation score and a cheap proxy label can come from surface evidence the two share rather than from the construct. Re-reading the proxy away from the span the score reads separates the two, with a pre-registered hold-out reported as undecidable.

2. When Residualization Helps an Audit: Format Effects, Slice Gains, and Their Limits.

Daein Weon, Dong Ho Kang.

arXiv:2609.24194, September 2026.

Subtracting a predictable surface component from reward-model and judge scores reduces a format effect while preserving correctness margins, but does not transfer predictably outside the edit template it was fitted on: an audit diagnostic, not a deployable score.

RESEARCH EXPERIENCE

Independent Research

2025–2026

Designed and ran the studies above end to end: benchmark construction, inference-time controllers for small language models, pre-registered analysis, and release of code and per-episode traces, all without model training.

PROFESSIONAL EXPERIENCE

Z-Order, Irvine, California — Solutions Engineer

Sep 2025 – Feb 2026

Analyzed customer-inquiry data to streamline the response workflow; edge-case testing contributed to 10+ critical bug fixes.

K-Spot, Irvine, California — Full-Stack Engineer

Jun 2025 – Feb 2026

Built a bilingual directory platform (React Native, Next.js, Firebase); led code reviews and the shared mobile/web frontend.

Transity, Irvine, California — Full-Stack Engineer

Mar 2025 – Jun 2025

Integrated subway data across 190+ cities; cleaned 620,000+ records and loaded 617,123 validated entries without downtime.

Kookmin University — Team Leader, AI Big Data Analysis project

Oct 2024 – Nov 2024

Led a team predicting trade volume from six years of economic data with CNN+LSTM models on 1,081 preprocessed entries.

Republic of Korea Army — Sergeant, Intelligence

Aug 2020 – Feb 2022

RESEARCH PRACTICE

Predictions and falsification conditions are registered in dated notes before each run, and results that fail them are reported as failures. Across this line of work three pre-registered predictions returned negative and two previously reported effects were withdrawn after control analyses — one an artifact of right-censoring, one a mis-classification of a benchmark’s labels.

TECHNICAL SKILLS

Languages Python, L^AT_EX

ML PyTorch, Hugging Face Transformers, vLLM, scikit-learn

Evaluation benchmark construction, LLM-as-judge protocols, bootstrap and permutation inference, pre-registration

SERVICE AND OTHER

Reviewing Reviewer, ICLR 2027 (reciprocal reviewing programme)